

PFD March 12

Critical Path Network

Also known as CPM. This is the longest path through a network chart where all activities have $ES = LS$ (Early Start = Late Start) and $EF = LF$ (Early Finish = Late Finish). All such activities are **CRITICAL**

Critical activities have 0 slack

PERT - Programme Evaluation and Review Technique

- The duration is probabilistic
- It is difficult to estimate exactly how long an activity will take. So, the inventors of pert decided to make use of 3 guesses at the duration
- From those 3 guesses they decide to calculate an average duration, which they decided to call Expected Duration

$$ED = A + 4B + C / 6$$

Predecessor		Optimistic(Known as A)	Most Likely (Known as B)	Pessimistic (Known as C)	ED	Celling()Ec
-	A	1	3	5	3	

- We use the ED figures on our network to calculate the Earliest Finish Time For the project. This figure is called MU(The greek letter)
- We need to calculate another figure for our network which is to be called s. But first, we calculate s squared
- To calculate s squared, we add up all the variance along the critical path
- A variance is calculated for each activity using the formula
- The probability of ending project at MU is 50%
- $Variance = (c-A/6)^2$
- $z = x - MU/s$
- We need to find the probability associated with z. To do so, we need to look up the probability on a table called a z-table
- Another name for the z table is the standardised normal distrubution